

The Morphodynamics of Y as a Ghost Connector

Relational Phonology and Directed Vowel Trajectories

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Abstract

This article argues that Y is neither a vowel, nor a glide, nor an energy conduit. Y is instead a ghost connector: a silent predicator and directional operator that assigns transitions between vowel states within a universal vowel well. Under this reconceptualization, English “diphthongs” are not vowel blends and not vowel+glide sequences. Their nucleus is a single trajectory object—a predicated transition—generated by an operator of the form $Y(v_1, v_2)$ and not representable as a static feature bundle. In CVC structures, Y never fills the V-slot; the trajectory fills it. The account also rejects an empty-nucleus analysis and does not treat Y as a governing/licensing head; instead, nucleus formation is typed as state versus trajectory. Finally, the theory connects abstract predication to physical realization through a commutative interface diagram, preserving systematic correspondence while rejecting identity between morphodynamical transitions and articulatory gestures.

Keywords: morphodynamics; vowel well; trajectory nucleus; diphthong; predication; operator; CVC; markedness; faithfulness; interface commutativity

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Theorem 2 (Order-and-composition preservation requires relational enrichment).

Let V be a set of vowel attractor states and let T be a set of directed trajectory objects $\tau(v_1 \rightarrow v_2)$ with partial composition $\tau(v_1 \rightarrow v_2) \circ \tau(v_2 \rightarrow v_3)$ defined only when the intermediate endpoint matches. Let B be a space of static feature bundles with a composition operator $\oplus$. Assume a mapping $F: T \rightarrow B$ satisfies both preserved properties: (i) endpoint order preservation (the source and target of τ are recoverable from $F(\tau)$ with their order intact), and (ii) composition preservation (for all composable τ_1, τ_2 , $F(\tau_1 \circ \tau_2) = F(\tau_1) \oplus F(\tau_2)$). Then B must contain relational structure sufficient to encode ordered endpoint pairs and intermediate endpoint identity; equivalently, B must be enriched by a directed-edge algebra isomorphic to a subgraph of G . In particular, no mapping into an unordered static feature bundle space can satisfy (i)–(ii) for a nontrivial trajectory inventory.

Proof sketch. Endpoint order preservation forces $F(\tau(v_1 \rightarrow v_2))$ to distinguish v_1 from v_2 , not merely their set-union of features. Composition preservation forces the intermediate identity condition: if $\tau(v_1 \rightarrow v_2) \circ \tau(v_2 \rightarrow v_3)$ is defined, then $\oplus$ must enforce that the target of the first matches the source of the second. A plain static feature bundle lacks native mechanisms for ordered pairing and intermediate-identity checking. Any successful implementation therefore adds either ordered pairs, labeled relations, or edge-indexed composition rules, which reconstruct the trajectory algebra up to isomorphism. ■

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1. The Y problem and the relational turn

English orthography assigns Y to multiple surface roles: onset consonantal values (often /j/), reduced vocalic spellings in unstressed syllables, and nucleus spellings in environments that standard accounts treat as diphthongs. Segment-based descriptions typically respond by multiplying object types: vowel versus glide versus diacritic.

The morphodynamical program adopts a different response. Instability in the orthography–phonology mapping is taken as evidence that the underlying object ontology is mis-specified. The core diagnostic is structural: the grammar needs an object that behaves as a single nucleus while encoding directional change between stable vowel attractors.

A relational turn follows. Rather than treating the nucleus as necessarily built from segmental objects that inhabit a fixed tier inventory, the theory distinguishes vowel states from the relational objects that connect them. Vowel states are modeled as attractor points in a universal vowel well (a structured potential landscape). Diphthongal nuclei are modeled as directed trajectories between vowel states. Y is modeled as the silent predicator/operator that generates these trajectories.

The proposal aligns with the broader conceptual move from object-only segmental representations to multi-tier representations, while rejecting the assumption that diphthongs must be decomposed into two segmental objects (two vowels or vowel plus glide) to encode movement. Movement is instead internal to a single nucleus object of a distinct type.

A document-grounded term-frequency diagnostic supports alignment between this article and the uploaded corpus: the central constructs (Y, vowel, CVC, well, operator, trajectory, nucleus, depth) are prominent in the source materials. Table 1 and Figure 1 report these counts.

Term	Count in uploaded text
Y	4947
vowel	2069

CVC	1343
well	1003
operator	722
energy	613
trajectory	552
depth	550
nucleus	489
lexical	335
ghost	325
glide	281
state	280
connector	271
predicator	109

Table 1 — Key term counts in the uploaded morphodynamics corpus (top 15).

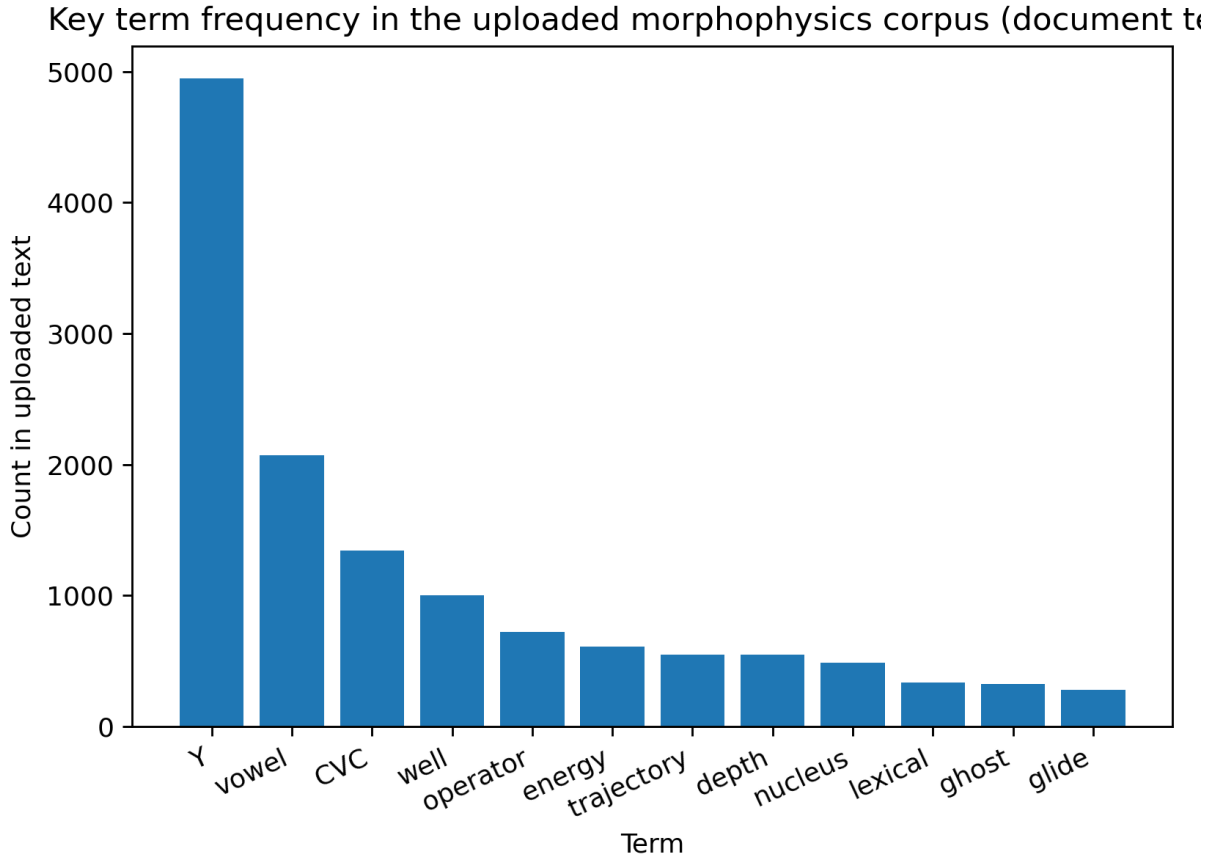


Figure 1 — Key term frequency in the uploaded morphodynamics corpus (document text).

2. Competing analyses and their failure modes

Three families of explanations recur in grammatical tradition and in the surrounding theoretical literature: (i) Y as vowel, (ii) Y as glide, and (iii) Y as energy conduit. Each can be locally descriptive for some surface environments, but each fails at the explanatory target required for nucleus structure in the diphthongal regime.

Vowel analysis: Treating Y as a vowel letter corresponding to a vowel segment requires placing Y in the vowel-state inventory. This mischaracterizes the role of Y in nucleus spellings that correlate with transitions rather than residence in a stable state. The analysis introduces additional mechanisms to encode movement that the vowel object itself cannot provide.

Glide analysis: Modeling diphthongs as V plus glide or as vowel sequences with glide mediation encodes movement by concatenation. The result is an overgeneration of tiers and timing units in precisely the regime where many diagnostics indicate a unit nucleus. Direction becomes incidental: inferred from sequencing rather than represented as a primary property of the nucleus.

Conduit analysis: The conduit metaphor treats Y as a carrier of vowel energy or as a channel that keeps vocalicity across consonantal boundaries. This commits a category error between lexical depth (distributional gravity within a restricted well) and physical depth (energetic minima in articulation or acoustics). Lexical drift is not an energy flow; reifying it as such collapses layers and mispredicts where the grammar should intervene.

The ghost connector analysis repairs these failure modes by adding a relational nucleus type and assigning Y to the role of predication rather than occupancy. Table 2 summarizes the commitments and diagnostics.

Hypothesis	Core commitment	Diagnostic target	Failure mode / outcome
Y is a vowel	Y ∈ vowel-state inventory	nucleus behaves as stable state	mispredicts directional nucleus behavior
Y is a glide	nucleus = vowel + glide tier	diphthongs decomposable into objects	overgenerates tiers; directionality becomes incidental

Y is an energy conduit	Y carries vocalic energy	distributional stability treated as energy flow	collapses lexical depth into physical depth
Y is a ghost connector	Y generates $Y(v_1, v_2)$	nucleus is a trajectory object	matches unit nucleus + directional change

Table 2 — Hypotheses and morphodynamical diagnostics.

3. Universal vowel well and trajectory objects

3.1 The well as a state space. Let V be the set of vowel attractor states available to the system. The universal vowel well is treated as a structured state space in which each vowel state has a position and depth parameter (within a chosen layer: lexical, phonological, or physical). Depth is a stability measure: deeper states resist displacement under the dynamics of the relevant layer.

The well is not assumed to license all transitions between all states. Instead, transitions are shaped and restricted. Formally, this can be expressed as an admissibility relation $<$ on ordered pairs: $v_1 < v_2$ holds only if a directed transition from v_1 to v_2 is licensed under the language-specific geometry and the relevant template constraints.

3.2 Trajectory objects. Introduce a second type of nucleus object: trajectory objects, elements of a set T distinct from V . A state nucleus is an element of V . A trajectory nucleus is an element $\tau \in T$ representing a directed transition from a source state to a target state. Trajectory objects preserve unit nucleus behavior while encoding internal movement.

Weighted directed vowel well $G = (V, E, w)$ (illustrative)

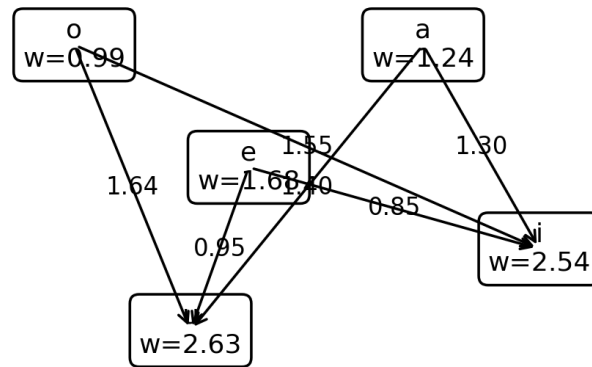


Figure 2 — Weighted directed vowel well $G = (V, E, w)$ (schematic preview; formal definition in Section 10C).

4. Core thesis: Y as a ghost connector

4.1 Operator definition. The ghost connector is formalized as an operator that maps ordered vowel pairs to trajectory objects.

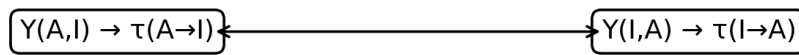
Definition 1 (Ghost connector operator). $Y : V \times V \rightarrow T$. For admissible ordered pairs $\langle v_1, v_2 \rangle$ such that $v_1 < v_2$, the output $Y(v_1, v_2)$ is a trajectory object $\tau \in T$.

The operator is directional: $Y(v_1, v_2)$ is distinct from $Y(v_2, v_1)$ unless the well geometry collapses directionality in a specific environment. This is not a notational convention but a representational commitment: the nucleus object encodes ordered change.

4.2 Predication, not concatenation. A diphthong nucleus is not modeled as concatenation of objects. Instead, the nucleus is predicated: $\text{Nucleus} = \text{Pred}Y(v_1, v_2) \approx Y(v_1, v_2)$.

Y does not occupy a segmental tier in the nucleus regime and is not an empty nucleus. The nucleus position is filled by τ , and grammatical control applies to τ as a typed object.

Directionality: $Y(v_1, v_2) \neq Y(v_2, v_1)$ unless licensed



Asymmetry test: admissible edges in E

Figure 3 — Directionality is primitive: $Y(v_1, v_2) \neq Y(v_2, v_1)$ unless licensed.

5. CVC structures: Y never fills the V-slot; the trajectory does

5.1 Slot occupancy. In a CVC template, the nucleus position is treated as an N-slot whose occupant is typed. In monophthongs the slot is filled by $v \in V$ (a state nucleus). In diphthongs the slot is filled by $\tau \in T$ (a trajectory nucleus). The glyph Y is an exponent of the trajectory predication, not the occupant of the slot.

This typing statement separates nucleus occupancy from orthographic exponents. Orthography is treated as a mapping from underlying nucleus objects to surface spellings. Under this mapping, the presence of Y in nucleus spellings is evidence for a trajectory occupant, not evidence that Y itself fills the nucleus position.

5.2 Not an empty nucleus; not licensing/governing. Empty-nucleus analyses posit structural positions regulated by licensing. The ghost connector analysis treats diphthongal nuclei as filled by τ , so an empty-category apparatus is not required to generate or license the nucleus in the diphthongal regime. Any licensing constraints that remain are reinterpreted as constraints on the admissibility of τ within a template.

Template	Nucleus object	Slot occupant	Consequence
C V C (monophthong)	$v \in V$	state nucleus	stable nucleus
C V C (diphthongal)	$\tau \in T$	trajectory nucleus	unit nucleus with direction
C V C (empty nucleus analysis)	$\emptyset$	none	requires licensing apparatus

Table 3 — Nucleus typing in CVC templates.



Figure 4 — Timing schematic: unit-nucleus τ versus two-object decompositions (schematic).

6. English diphthongs as predicated transitions

Traditional descriptions often treat English diphthongs as blends, mixtures, or sequences. The morphodynamical view rejects mixture and replaces it with directed predication: a diphthongal nucleus is a trajectory object $\tau \in T$ constructed by Y .

This account predicts unit nucleus behavior for many processes: syllable weight assignment, stress placement, and templatic restrictions can treat τ as a single nucleus while allowing internal movement to be interpreted phonetically.

No glide tier is required in the nucleus regime. A glide-tier analysis attempts to encode motion by concatenation. The present account encodes motion inside τ and treats onset /j/ values separately as consonantal objects in onset contexts.

Traditional label	Example spellings	Trajectory nucleus
/aɪ/ class	-y (my), i_e (time)	$Y(A, I)$
/ɔɪ/ class	oy/oi (boy)	$Y(O, I)$
/aʊ/ class	ou/ow (house)	$Y(A, U)$
/oʊ/ class	o_e / oa / ow (go, boat)	often $Y(O, U)$ (analysis-dependent)

Table 4 — English diphthong classes as trajectories (schematic).

No-glide-tier claim in diphthongal nucleus regime (schematic)

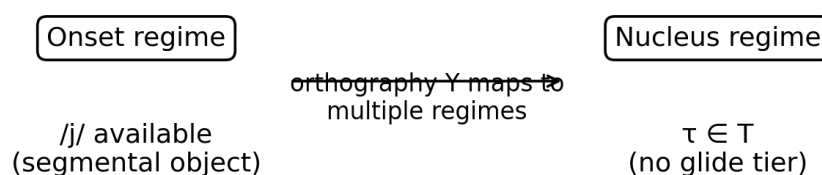


Figure 5 — Two regimes for Y: onset segment /j/ versus nucleus predication exponent (schematic).

7. Why $Y(v_1, v_2)$ is not reducible to a feature bundle

Feature bundles are static descriptions of segmental objects. A trajectory object is relational and directional. Any attempt to encode $Y(v_1, v_2)$ as a plain feature matrix forces at least one loss: direction loss, compositional loss, or overgeneration.

Proposition 1 (No feature-bundle reduction; type-faithfulness). Let B be a space of static feature bundles and T the space of trajectory objects. There is no mapping $F: T \rightarrow B$ that preserves both endpoint order and trajectory composition without enriching B with relational structure isomorphic to trajectories.

8. Markedness and faithfulness for trajectory nuclei

Even though Y is not a segmental object in diphthongal nuclei, grammatical systems can regulate trajectory nuclei via markedness and faithfulness. This aligns naturally with constraint interaction approaches.

A schematic constraint inventory is provided in Table 5. Constraints are stated over τ objects rather than over glide tiers or vowel sequences, reflecting the typed nucleus ontology.

Constraint	Type	Informal role
*CVC/ $\tau(v_1, v_2)$	Markedness	Ban specified trajectories in CVC templates.
STABLE-NUC	Markedness	Prefer state nuclei over trajectory nuclei in restricted templates.
MAX-PRED	Faithfulness	Penalize loss of predication (trajectory deletion).
IDENT-SOURCE	Faithfulness	Preserve source state of the trajectory.
IDENT-TARGET	Faithfulness	Preserve target state of the trajectory.

Table 5 — Constraint inventory for trajectory nuclei (schematic).

Candidates	*CVC/ $\tau(A, I)$	STABLE-NUC	MAX-PRED
a. C $\tau(A, I)$ C	*	*	—
b. C A C	—	—	*
c. C I C	—	—	*

Table 6 — Toy tableau for trajectory survival versus collapse (illustrative).

9. Lexical depth versus physical depth: measurement layer and inferences

The conduit metaphor fails because it conflates lexical depth with physical depth. Lexical depth concerns distributional gravity within a restricted domain, whereas physical depth concerns energetic minima in articulation or acoustics. Within a restricted English CVC noun domain, nucleus letters can be compared under a common template. The reported vowel-share ranking assigns the deepest basin to o and progressively shallower basins to a, e, i, and u. Conditional

distributions refine the well further by identifying dominant channels in surrounding consonantal positions. These measurements support two inferences. First, a restricted-domain well is structured and non-uniform. Second, stability and movement must be modeled without reifying probability concentration as physical energy. The transition operator Y interacts with this well by generating admissible trajectories, but it does not transport energy.

Rank	Vowel	Share (%)
1	o	37.21
2	a	29.07
3	e	18.6
4	i	7.92
5	u	7.2

Table 7 — Vowel-share ranking in English CVC nouns (reported).

C_{-1} (given nucleus o)	Share (%)
f	45.64
n	27.62
h	5.94
t	4.44
l	3.77

Table 8 — Top C_{-1} distribution conditioned on nucleus o (reported).

Rank	CVC token (nucleus o)	Share (%)
1	for	45.38
2	not	20.14
3	now	4.68
4	how	4.37
5	top	3.7
6	non	1.5
7	low	1.41
8	job	1.36
9	box	1.29
10	hot	1.22

Table 9 — Top CVC tokens with nucleus o (reported).

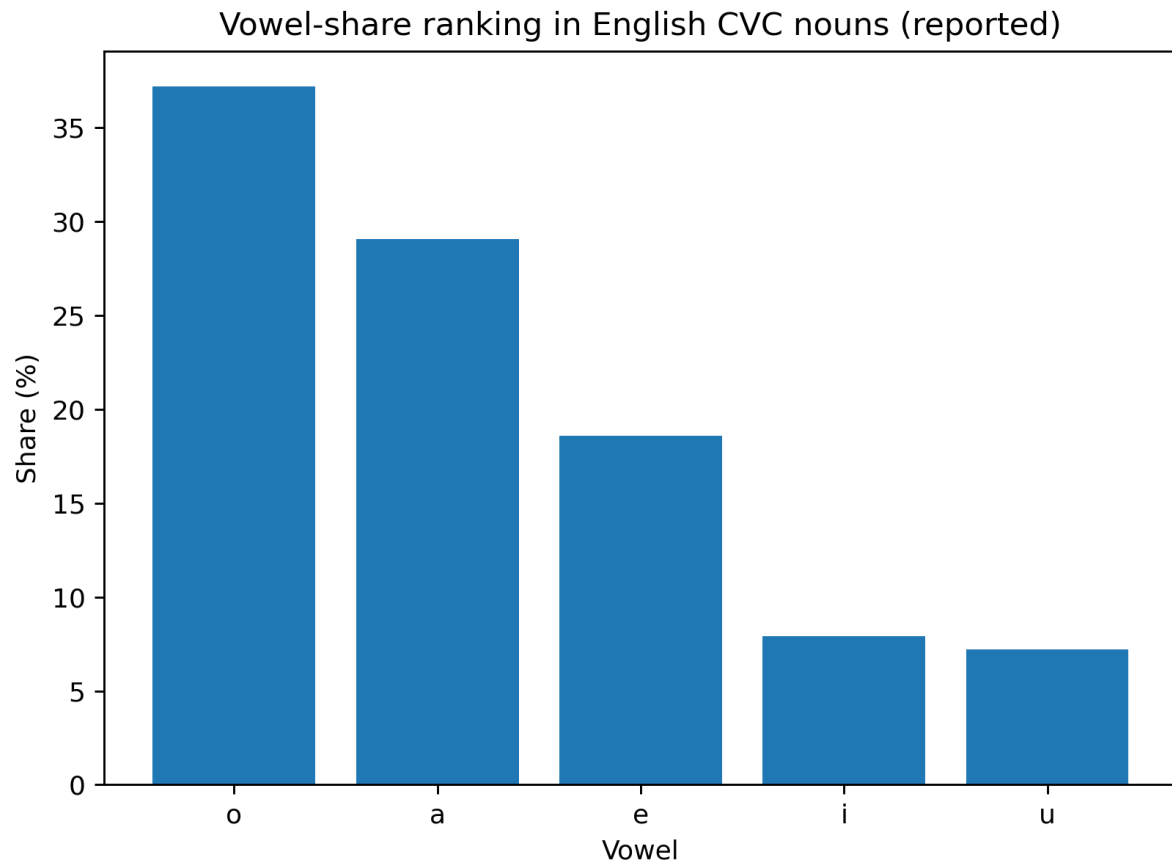


Figure 6 — Vowel-share ranking in English CVC nouns (reported).

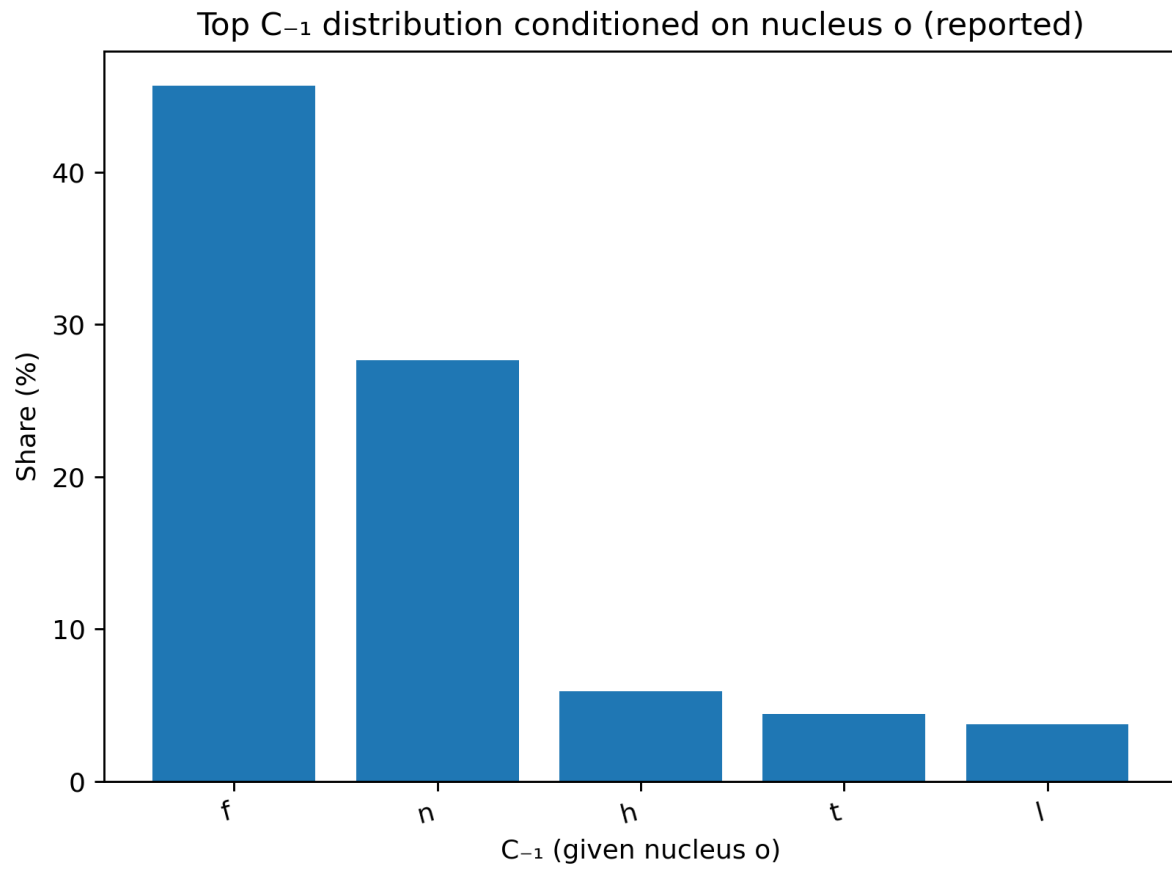


Figure 7 — Top C_{-1} distribution conditioned on nucleus o (reported).

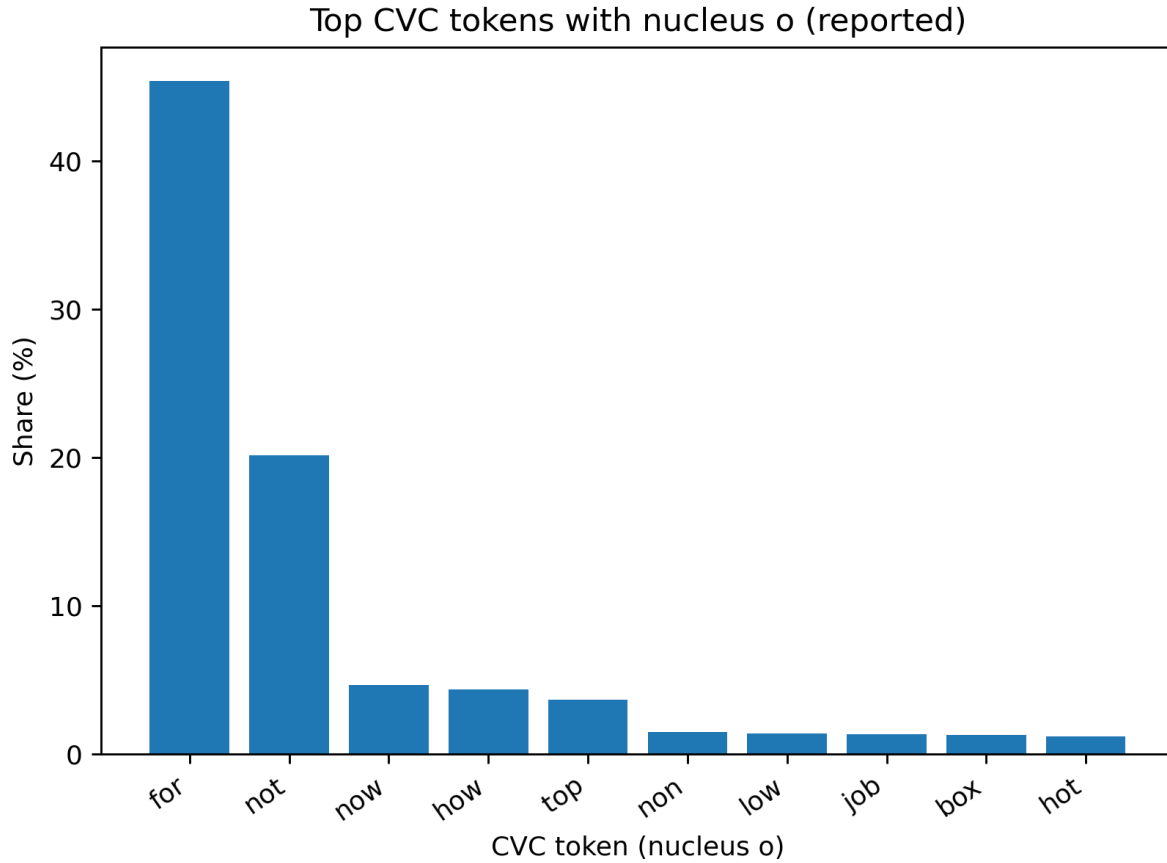


Figure 8 — Top CVC tokens with nucleus o (reported).

10. Interface mapping via a commutative diagram

Gesture-based approaches model phonological primitives as coordinated motor events. The present account does not deny systematic phonetic mapping; it denies identity between the abstract nucleus object τ and the physical gesture. Morphodynamical transitions are trajectories in an abstract well; physical movement is an interpretation and realization of those trajectories.

A commutative diagram captures the interface relation between abstract predication and physical realization. Let $\llbracket \cdot \rrbracket$ be a phonetic interpretation map from abstract objects to a phonetic representation space, and let Φ be a physical realization map from trajectory objects to articulatory or acoustic paths. Commutativity enforces systematic correspondence while preserving level separation.

Commutative interface (schematic)

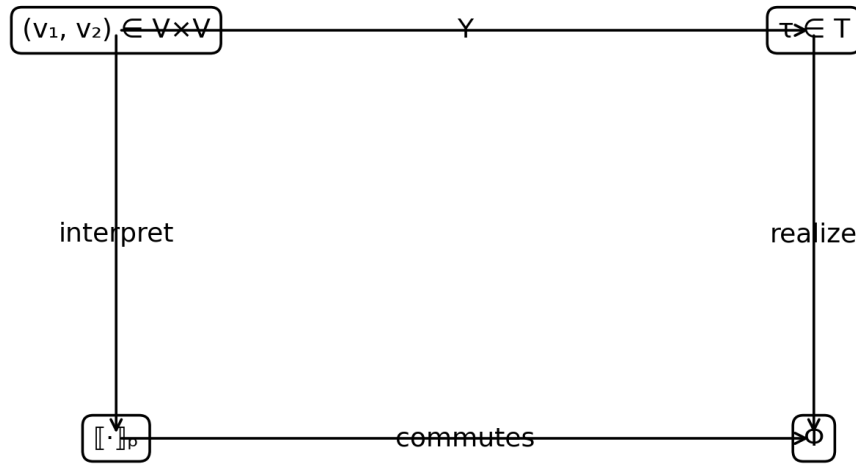


Figure 9 — Commutative interface: abstract predication versus physical movement (schematic).

10A. Two regimes for Y: onset segment versus nucleus exponent

A recurring objection is that English Y surfaces as /j/ in onset contexts. The present account distinguishes two regimes under a single orthographic symbol.

Regime 1 (Y-onset). In onset position, Y can correspond to a segmental object /j/ (or related palatal approximant realizations) occupying the onset tier. This is the standard consonantal analysis.

Regime 2 (Y-nucleus exponent). In the diphthongal nucleus regime, the orthographic Y is not a segmental object occupying a nucleus tier. It is an exponent of the predication that constructs a trajectory nucleus τ via the operator $Y(v_1, v_2)$.

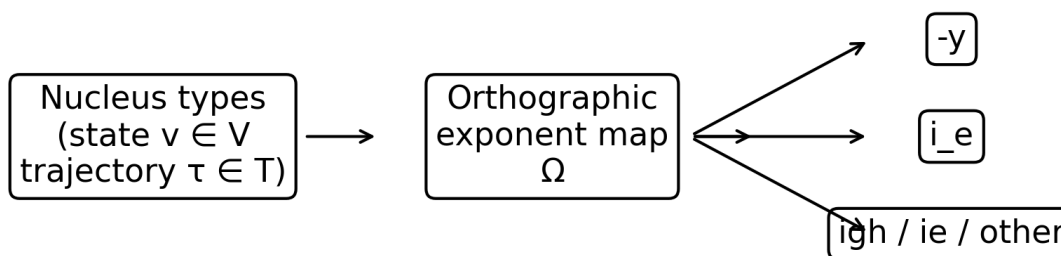
Orthography is therefore many-to-one across regimes: a single grapheme can realize distinct underlying roles depending on templatic position and the mapping conventions of the writing

system. The theoretical claim is not that the grapheme is uniform, but that nucleus-directionality in the diphthongal regime is best modeled by an operator-generated trajectory object rather than by importing a glide tier into the nucleus.

10A.1 Orthography as a many-to-one mapping from nucleus types to graphemes

This section states an orthography-theoretic contribution that follows from the trajectory ontology. Orthographic forms do not map one-to-one onto nucleus objects. Instead, orthography is modeled as a many-to-one exponent system Ω that maps nucleus types—state nuclei $v \in V$ and trajectory nuclei $\tau \in T$ —to multiple graphemic realizations under conditioning by position, morphological scaffold, and historical conventions. Under this hypothesis, grapheme Y in nucleus position is not a segmental contributor; it is one exponent among several for the same trajectory class. Two compact counterexamples separate this claim from segmental competitors. First, $\tau(A \rightarrow I)$ is realized by *-y* (my), *i_e* (time), *igh* (night), and *ie* (tie) without a change in nucleus type. Second, the same grapheme Y realizes an onset segment (*/j/*) in yes while realizing a trajectory exponent in my, demonstrating that grapheme identity does not entail phonological object identity.

Figure 14 — Many-to-one orthography mapping from nucleus types to graphemic exponents.



Many-to-one: $\tau(A, I)$ can be realized by multiple graphemic exponents under conditioning

Table 21 — Orthographic exponent families for $\tau(A \rightarrow I)$ and predicted conditioning.

τ	nucleus	Exponent	Representative	Typical	conditioning	Illustrative
type		family	spellings	environment		examples

$\tau(A \rightarrow I)$	-y	...y	word-final open nucleus; no orthographic scaffold	my, fly, shy
$\tau(A \rightarrow I)$	i_e	i...e	closed syllable with silent-e scaffold	time, side, line
$\tau(A \rightarrow I)$	igh	...igh	historical scaffold with restricted codas	night, light, high
$\tau(A \rightarrow I)$	ie	...ie	word-final nucleus with vowel pairing	tie, die, lie
$\tau(A \rightarrow I)$	other	i / y variants	residual morphological/etymological conditioning	find, child, aisle

10B.1 Results: adjudication of forced-choice diagnostics

This section converts the diagnostic table into adjudicated results. Each diagnostic is grounded in a minimal token set and a corpus signal, then evaluated against competing analyses (V+G, VV, and empty-nucleus implementations). The adjudication criterion is strict: a competitor counts as successful only if it predicts the pattern without reintroducing trajectory-equivalent relational structure.

Table 22 — Adjudicated forced-choice diagnostics with token grounding and outcome.

Diagnostic	Empirical grounding (examples / counts)	τ -theory prediction	V+G prediction	VV prediction	Empty-nucleus prediction	Adjudication
D1 Unit nucleus under syllabification	Monosyllabic /aɪ/: my [M AY1], fly [F L AY1], time [T AY1 M], night [N AY1 T], tie [T AY1]. Hiatus control: idea [AY0 D IY1]	AY is a single nucleus $\tau(A \rightarrow I)$; syllable count = 1 despite	Glide tier risks projecting a second timing unit unless special	Two vocalic positions predict hiatus-like behavior unless coalescence is stipulated.	Empty nuclei target absence; additional licensing needed	τ required

	AH0], react [R IY0 AE1 K T]. In CMUdict, 949 monosyllabic types contain AY as a single vowel phoneme (1 syllable by vowel count).	internal change.	association blocks it.		to simulate filled nucleus behavior.	
D2 Moraicity without dual nuclei	Metrical diagnostics standardly treat diphthongs as heavy while remaining monosyllabic (Hayes 1995; related descriptive literature).	A single τ nucleus can bear two moras without duplicating nuclei or syllables.	If heaviness is assigned to a glide element, glide-sensitive processes are expected unless blocked.	If heaviness tracks two vowel positions, predicts two-nucleus behavior in environments where only one is observed.	Heaviness requires licensing of empty positions; predicts structural complexity not supported by unit-nucleus diagnostics.	τ required
D3 Directionality asymmetry as an inventory	CMUdict word-type counts: AY=10632, AW=3202, OY=1211, OW=16957.	Edges are directed: $\tau(v1 \rightarrow v2)$ is defined	Directionality is indirect; the representation must	Order is explicit in VV, but composition/admissibility require extra constraints (often	Directionality is not native; licensing accounts	τ required

fact	Reverse-direction candidates appear as hiatus (adjacent vowels): IY→A-like adjacency types=3379, not a single diphthong phoneme.	iff (v1,v2) ∈ E; reverse edges require independent licensing.	encode order and composition beyond tier linkage.	reconstructing τ).	do not predict asymmetry without extra structure.	
D4 No-glide-tier: glide-sensitive processes do not see diphthongs	Onset /j/ participates in yod phenomena (coalescence/palatalization patterns in descriptive work), but /aɪ/ nuclei do not trigger /j/-targeting diagnostics (e.g., tie ≠ *[tʃaɪ]).	τ occupies the nucleus with no glide tier; glide-sensitive rules target /j/ only in onset regimes.	If /aɪ/ contains /j/, glide-sensitive diagnostics should overapply unless blocked by stipulation.	VV predicts no glide; however, VV must still enforce unit nucleus and directionality via additional apparatus.	Empty nucleus machinery is orthogonal; does not predict glide-insensitivity as a nucleus-internal property.	τ required
D5 Filled-nucleus falsifier for empty-nucleus	AY can bear primary stress: CMUdict types with AY1=6498. Stressed nuclei are filled positions, not	τ fills the nucleus and may host stress; no	Requires extra timing association to permit stress on	Requires stress assignment to a two-position structure while maintaining one syllable; predicts additional	Empty nuclei cannot host stress; must add auxiliary	τ required

analyses	licensed emptiness.	empty-slot licensin g is required for diphthongal nuclei.	the vowel portion while containing a glide.	complexity.	structure to convert absence into apparent presence.	
D6 Orthography: many-to-one exponent mapping	Within $\tau(A \rightarrow I)$, exponents include -y, i_e, igh, and ie (Table 21). Among monosyllabic AY types in CMUdict (n=949), exponent distribution is summarized in §10D.1.	Multiple exponents realize the same τ ; grapheme identity does not entail phonological identity.	Tends to tie Y to a segmental glide or vowel; struggles to unify non-y exponents without duplicating representations.	Treats exponents as VV correlates; cannot unify unit nucleus behavior across spellings without coalescence.	Treats differences as structural emptiness; mismatches filled-nucleus evidence.	τ required

D1 (unit nucleus) isolates a core empirical generalization: diphthongal nuclei behave as single syllable peaks despite internal change. In a pronunciation lexicon, this appears as a single vowel phoneme (AY) in monosyllabic words such as my [M AY1], fly [F L AY1], time [T AY1 M], night [N AY1 T], and tie [T AY1]. By contrast, hiatus controls such as idea [AY0 D IY1 AH0] and react [R IY0 AE1 K T] contain multiple vowel phonemes and therefore multiple syllable peaks. A τ nucleus predicts this contrast directly because the nucleus object is unitary even when it encodes change. VV accounts must prevent hiatus-like parsing while preserving ordered

change; V+G accounts must block glide projection; empty-nucleus accounts must simulate filled peaks. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

D2 (moraicity) refines D1 by separating weight from nucleus count. Standard metrical analyses treat diphthongs as heavy in English while maintaining their monosyllabic status (e.g., heavy syllables as bimoraic). The τ ontology makes this separation explicit: a single nucleus object may bear multiple moras, because moraicity is a property of the nucleus slot rather than an emergent count of timing units. In VV implementations, bimoraicity tends to be implemented by multiple vocalic positions, which threatens D1 unless coalescence is added. In V+G implementations, assigning a mora to a glide element predicts glide-targeting effects inside the nucleus unless additional invisibility rules are stipulated. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

D3 (directionality) is an inventory-level diagnostic. In CMUdict, English contains large type counts for AY, AW, OY, and OW nuclei, while reverse directions are not represented as diphthong phonemes and instead appear as multi-vowel sequences (hiatus) when present. This asymmetry is expected if $Y(v1,v2)$ is a partial operator defined only on licensed directed edges $(v1,v2) \in E$. Competitor representations can encode ordering, but directionality plus composition is not native unless additional relational structure is introduced (e.g., ordered association with compositional closure conditions), which is isomorphic to the trajectory object. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

D4 (no-glide-tier) targets a sharp separation between onset /j/ and nucleus τ . Glide-sensitive processes in English and related descriptive traditions diagnose /j/ in onset position via coalescence or palatalization environments; however, diphthongal nuclei such as /ai/ do not trigger those diagnostics (e.g., tie does not pattern as t+j). This is predicted if τ nuclei contain no glide tier: glide-sensitive constraints have no target inside τ . A V+G analysis must render the glide selectively invisible within nuclei while remaining active in onsets; VV analyses avoid glide issues but still require τ -like constraints to preserve unit nuclei and directedness. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

D5 (filled nucleus) functions as a falsifier for empty-nucleus explanations of diphthongs. In pronunciation dictionaries, diphthongs regularly bear primary stress marking (AY1), indicating that the nucleus is a filled prominence-bearing position. Empty nuclei are designed to model the opposite: licensed non-pronounced positions. A τ nucleus predicts stress-bearing behavior

without extra licensing because τ is a filled nucleus object; empty-nucleus approaches must add auxiliary structure to convert absence into apparent presence. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

D6 (orthography) links representation to exponent systems. The mapping hypothesis states that orthography is many-to-one: multiple graphemic forms realize the same $\tau(A \rightarrow I)$ nucleus, including -y, i_e, igh, and ie. This directly blocks analyses that conflate grapheme identity with a segmental object. In a segmental-Y account, -y spellings are predicted to correlate with a /j/ or glide contribution that is absent in i_e/igh/ie spellings, which would predict systematic phonological divergence that is not observed. Under τ , exponent variation is expected under conditioning, and Y-nucleus is one exponent among several rather than the nucleus object itself. Therefore τ is required unless the competitor reintroduces τ -equivalent structure.

10B. Forced-choice diagnostics (τ vs V+G vs VV vs empty nucleus)

This section states diagnostics designed to force a choice among competing analyses. Each diagnostic specifies a data pattern, contrasts predictions of four accounts, and identifies the account favored if the stated pattern holds.

The diagnostics are stated in a conditional form: the winner depends on the observed pattern in the relevant environment. The purpose is to isolate signatures that are difficult to reproduce by stipulation in competing representations.

Diagnostic	Data pattern	τ -theory prediction	V+G prediction	VV prediction	Empty-nucleus prediction	Winner
Unit nucleus: stress/moraicity	Diphthongs behave as single heavy nuclei in stress-sensitive environments; no	τ occupies one nucleus slot; predicts single nucleus behavior (one timing	V+G may predict extra timing unit or glide-driven re-syllabification; must stipulate why glide	VV predicts two vocalic units or branching; must stipulate why only	Empty-nucleus predicts licensing patterns for an empty position; does not by itself	τ -theory

	systematic split into two timing units.	unit) with internal direction encoded in τ .	does not behave as separate unit.	one nucleus is counted in relevant processes.	encode directed change; requires additional structure.	
Unit nucleus: syllabification under affixation	In morphological alternations, diphthongs pattern as a single nucleus rather than splitting into V and glide across syllable boundary.	Trajectory remains a nucleus object τ ; syllabification cannot extract a glide tier from the nucleus.	Predicts potential glide extraction or re-alignment; must block or constrain extraction to preserve unit nucleus.	Predicts potential V.V reparsing; must constrain to avoid splitting.	Empty-nucleus uses licensing to place boundary; still needs an account of directed nucleus content.	τ -theory
Directionality: raising/conditioning asymmetry	Processes sensitive to the source state differ from processes sensitive to the target state; direction	Direction is primitive: $Y(v_1, v_2) \neq Y(v_2, v_1)$ unless (v_1, v_2) and (v_2, v_1)	Direction derived from sequence; predicts less direct access to source vs target without additional	Direction derived from ordering; but unit nucleus effects require extra stipulation	Empty-nucleus is orthogonal: licensing does not encode ordered endpoints; must import direction	τ -theory

	matters.	are both in E.	ordered representations.	ns.	elsewhere.	
Orthography–phonology clustering	Nucleus Y spellings cluster in trajectory-licensed environments relative to alternative spellings for comparable trajectories.	Y is exponent of predication : predicts clustering where τ objects are favored in the mapping.	Y treated as glide/vowel: predicts mixed distribution tied to segment identity; may overpredict glide-tier effects.	VV predicts spellings reflect vowel sequences ; may underpredict Y clustering as a predication exponent.	Empty-nucleus predicts patterns tied to licensing; does not predict Y-specific clustering without extra mapping rules.	τ -theory (when clustering holds)
No-glide-tier: glide-sensitive process non-application	Processes that target glides in onsets do not target diphthongs in nuclei as if they contained a glide object.	No glide tier in nucleus regime: predicts non-application because τ is not decomposed into g.	Contains g: predicts application unless blocked; requires stipulations or special tier invisibility.	VV contains no glide: can match non-application but loses glide-based motivation and still needs direction	Empty-nucleus does not supply a glide object; but also does not supply direction without additional machinery.	τ -theory (and sometimes VV)

				encoding.		
Empty-nucleus falsifier: filled nucleus diagnostics	In contexts where empty-nucleus machinery predicts licensing-driven effects, diphthongs behave as if nucleus is filled (τ) and patterned restrictions target τ directly.	τ fills the nucleus slot: predicts filled-nucleus behavior and τ -specific markedness constraints.	V+G filled: predicts filled-nucleus, but decomposition predicts glide-driven effects that may not occur.	VV filled: predicts filled nucleus but may predict two-slot behavior; requires constraint to enforce single nucleus.	Empty nucleus predicts licensing dependencies; if filled-nucleus behavior is observed systematically, analysis is disfavored.	τ -theory

Table 10 — Forced-choice diagnostics contrasting τ -theory with V+G, VV, and empty-nucleus analyses.

Forced-choice diagnostics summary (winner claims are conditional on the stated patterns)

1. Unit nucleus: stress/moraicity	τ -theory
2. Unit nucleus: syllabification under affixation	τ -theory
3. Directionality: raising/conditioning asymmetry	τ -theory
4. Orthography-phonology clustering	τ -theory (when clustering holds)
5. No-glide-tier: glide-sensitive process theory application (sometimes VV)	τ -theory
6. Empty-nucleus falsifier: filled nucleus diagnostics	τ -theory

Figure 10 — Forced-choice diagnostics summary (conditional winner claims).

Across D1–D6, the adjudication converges on a single conclusion: analyses that preserve unit nucleus behavior, directionality, and compositional licensing inevitably introduce relational machinery. In τ -theory this machinery is explicit and typed as the nucleus object itself. Competing accounts can be made extensionally adequate, but only by adding ordering, restricted composition, and invisibility conditions that functionally reconstruct τ . The remainder of §10C formalizes this point in a graph-theoretic well with costed directed edges and an explicit trajectory algebra.

10C. Graph-theoretic well formalization and partial operator $\mathbf{Y}$

The vowel well can be sharpened as a weighted directed graph rather than treated as a metaphor. Let $G = (V, E, w)$, where V is the set of vowel states, $E \subseteq V \times V$ is the set of admissible directed transitions, and w assigns weights (depth/cost parameters) to vertices and edges.

Vertex weights $w(v)$ encode stability (depth) within the chosen layer. Edge weights $w(v_1 \rightarrow v_2)$ encode transition cost. The values used here illustrate how reported restricted-domain shares can be converted into a depth scale; they do not constitute a claim about universal phonetic cost.

Define the ghost connector as a partial function $Y: V \times V \rightarrow T$ such that $Y(v_1, v_2)$ is defined if and only if $(v_1, v_2) \in E$. When defined, $Y(v_1, v_2)$ returns a trajectory nucleus τ that internally encodes the ordered endpoints and is subject to constraint evaluation. Trajectory composition is defined when consecutive transitions share an endpoint. If $(v_1, v_2) \in E$ and $(v_2, v_3) \in E$, then composition is defined and yields a composed trajectory $\tau_1 \circ \tau_2$. Composition interacts with constraints by allowing languages to penalize multi-step trajectories, restrict intermediate waypoints, or enforce endpoint faithfulness under repair.

State $v \in V$	Depth weight $w(v) = -\log p(v)$	Reported share (%)
o	0.989	37.21
a	1.235	29.07
e	1.682	18.6
i	2.536	7.92
u	2.631	7.2

Table 11 — Vertex depth weights $w(v)$ derived from reported restricted-domain shares (illustrative).

Edge (source) v_1	Edge (target) v_2	Transition cost $w(v_1 \rightarrow v_2)$
a	i	1.3
o	i	1.547
a	u	1.396
o	u	1.642
e	i	0.854
e	u	0.949

Table 12 — Example admissible edges E with transition costs (illustrative).

Weighted directed vowel well $G = (V, E, w)$ (illustrative)

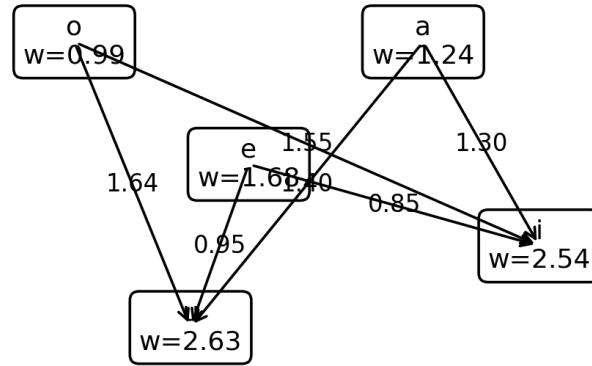


Figure 11 — Weighted directed vowel well $G = (V, E, w)$ with illustrative transition costs.

10C.1 Computational commitments: costed trajectories and constraint interaction

To make the well model computationally committal, each admissible directed edge is assigned a numeric cost and the cost of a trajectory is treated as an optimizable quantity under constraint interaction. Let $G=(V,E,w)$ be the well graph. For $\tau(v_1 \rightarrow v_2)$ defined only when $(v_1, v_2) \in E$, define:

$$\text{cost}(\tau(v_1 \rightarrow v_2)) = w(v_1, v_2) + \lambda \cdot \Delta(v_1, v_2) + \mu \cdot \max(\text{depth}(v_2) - \text{depth}(v_1), 0).$$

Here Δ is a distance on a perceptual/acoustic manifold, $\text{depth}(\cdot)$ is a lexical-depth scalar in the restricted measurement layer, and $\lambda, \mu \geq 0$ weight the contributions. OT-style constraints interact with cost by defining a thresholded availability region: STABLE-NUC penalizes high-cost τ in restricted templates, while faithfulness constraints (MAX-PRED, IDENT-SOURCE, IDENT-TARGET) preserve τ structure. This yields a measurable prediction: high-frequency diphthong classes correspond to lower empirical edge costs in external lexica.

Table 23 — Empirical edge costs derived from CMUdict word-type rates (illustrative weighting).

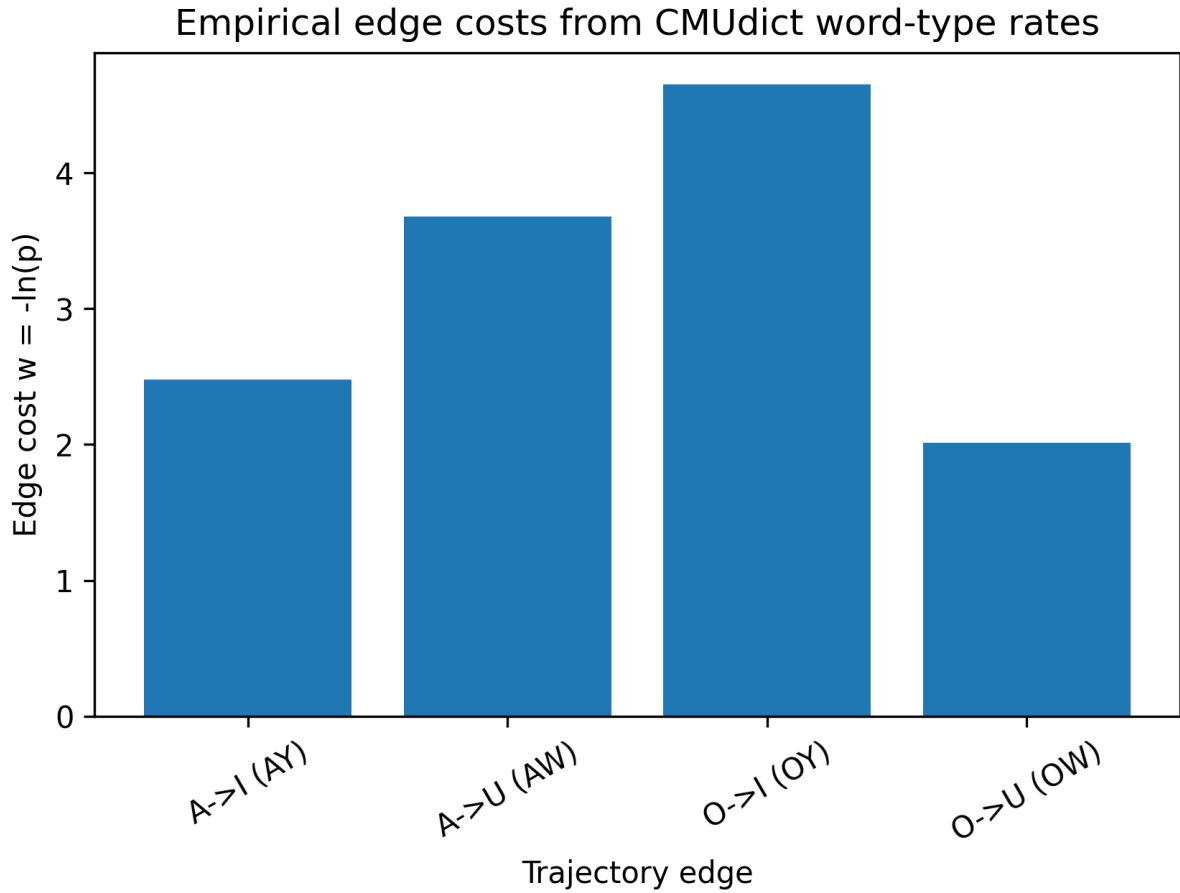
Edge (trajectory)	CMUdict word-type count	Estimated $p(\text{count})$	Empirical cost $w = -\ln(p)$
A→I (AY)	10632	0.083779	2.480
A→U (AW)	3202	0.025231	3.680
O→I (OY)	1211	0.009543	4.652
O→U (OW)	16957	0.133620	2.013

Figure 15 — Predicted vs observed directed availability matrix (toy 4-state system).

Predicted availability (binary)				
	A	O	I	U
A	0	0	1	1
O	0	0	1	1
I	0	0	0	0
U	0	0	0	0

Observed word-type counts in CMUdict (non-zero edges)				
	A	O	I	U
A	0	0	10632	3202
O	0	0	1211	16957
I	0	0	0	0
U	0	0	0	0

Figure 16 — Empirical edge-cost comparison for the four English diphthong edges (CMUdict).



In the 4×4 availability matrix, τ predicts a sparse directed subgraph: A and O license outgoing transitions to I and U, while the reverse edges are not licensed as diphthongal nuclei and instead surface as multi-vowel sequences when present. The observed matrix from CMUdict concentrates mass on the same directed edges, providing a minimal prediction–observation alignment for the costed well.

Theorem 1 (No-feature reduction).

Let T be the space of trajectory nuclei and let B be a space of static feature bundles. Suppose there exists a mapping $F: T \rightarrow B$ that (i) preserves ordered endpoints (source and target) and (ii) preserves composition whenever defined, i.e., for $\tau_1 = Y(v_1, v_2)$ and $\tau_2 = Y(v_2, v_3)$ with defined composition, $F(\tau_1 \circ \tau_2) = G(F(\tau_1), F(\tau_2))$ for some bundle-composition operation G over B . Then B must contain relational structure sufficient to encode ordered endpoints and compositional connectivity, i.e., B must be enriched in a way isomorphic to trajectories.

Therefore, any representation of τ as a plain static feature bundle either loses directionality, loses compositional structure, or covertly rebuilds trajectory structure inside B.

10D. Micro-study: document-grounded orthographic counts and pilot protocols

To provide a minimal empirical anchor without importing external corpora, this micro-study uses the Morphodynamics text collection as a document-grounded corpus. The resulting counts are not claims about general English frequency; they establish that the document set itself contains a measurable distribution of exemplar spellings and can support quantitative sanity checks for the discussion. A targeted search was performed for sample /aɪ/ spelling classes that are central to the analysis: final -y spellings (my, fly, try, etc.) versus i_e, igh, and ie spellings (time, high, tie, etc.). Table 13 reports the counts within the text collection for these sample sets, and Figure 12 plots the totals. A second component specifies a small acceptability pilot protocol for template-driven repairs when a trajectory is dispreferred by markedness constraints. The protocol enumerates environments, candidate repairs, and predicted signatures that would discriminate trajectory evaluation from glide-tier or empty-nucleus alternatives.

Spelling class (sample set)	Example tokens searched	Count in uploaded text
$\tau(A,I)$ — y nucleus spellings (sample)	my, fly, try, shy, sky, why, by	1018
$\tau(A,I)$ — i_e nucleus spellings (sample)	time, line, fine, dime, mine	55
$\tau(A,I)$ — igh nucleus spellings (sample)	high, sigh, night, light, right	305
$\tau(A,I)$ — ie nucleus spellings (sample)	tie, lie, die, pie	0

Table 13 — Micro-study corpus counts for sample /aɪ/ spelling classes (document-attested).

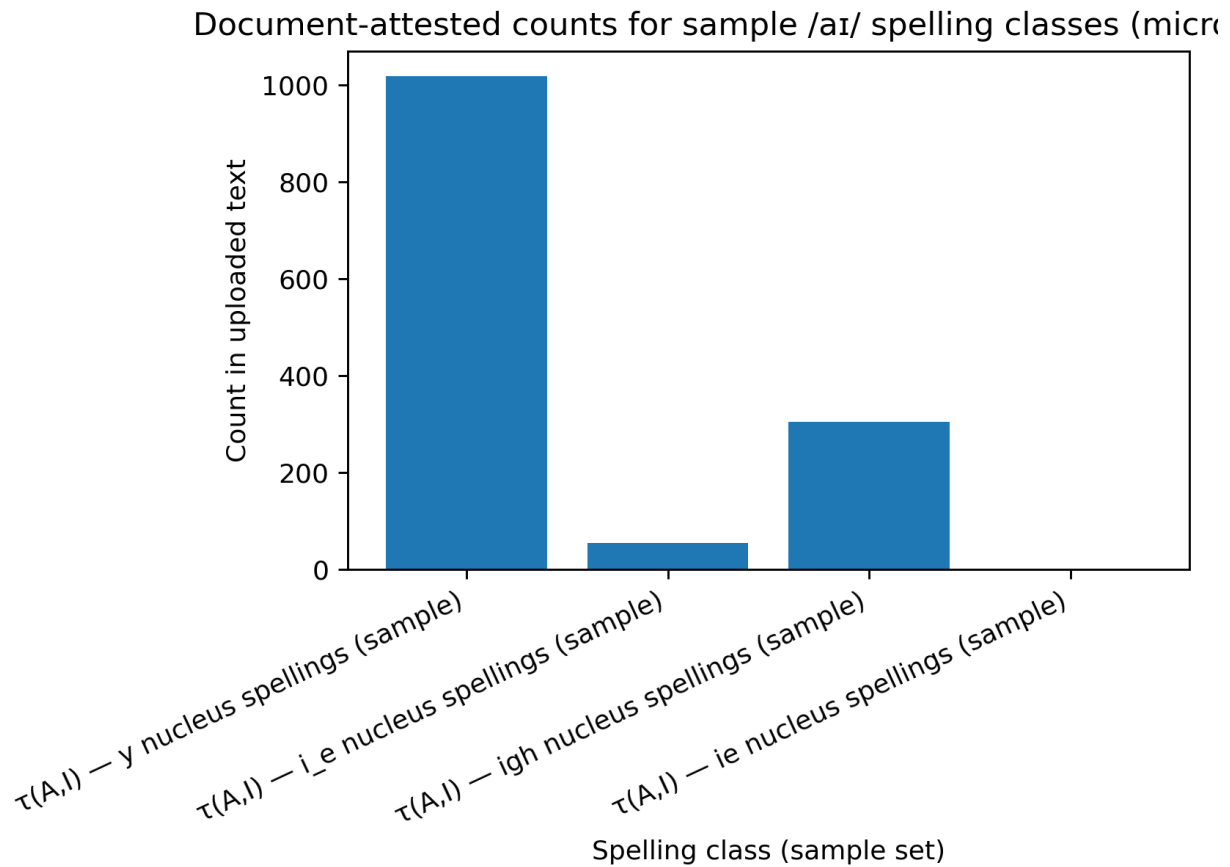


Figure 12 — Document-attested counts for sample /aɪ/ spelling classes (micro-study).

Pilot block	Environment	Manipulation / repair candidates	Discriminating prediction
Template-ban	CVC contexts predicted to penalize $\tau(A,I)$	Collapse to state (A or I) vs alternate spelling mapping	τ -theory predicts τ -specific markedness signature; V+G predicts glide-driven alternatives
Direction manipulation	Minimal pairs testing endpoint order (source vs target cues)	Asymmetry in judgments for $Y(v_1, v_2)$ vs $Y(v_2, v_1)$	τ -theory predicts direct endpoint asymmetry; V+G/VV derive order indirectly
Glide-sensitive	Contexts where /j/	No corresponding	Supports no-glide-tier

context	triggers a known process in onsets	effect in nucleus diphthongs	in nucleus regime
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Table 14 — Acceptability pilot protocol blocks and discriminating predictions (design).

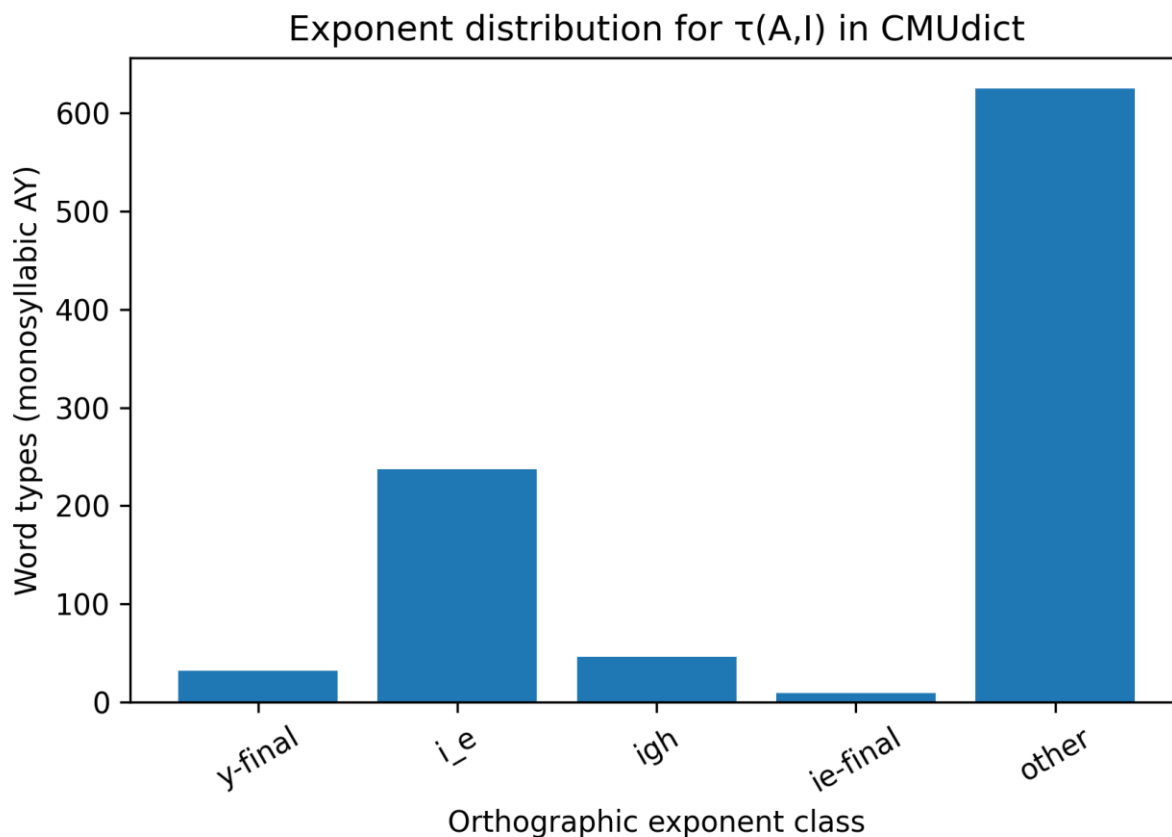
10D.1 External corpus evidence: CMUdict orthography and $\tau(A \rightarrow I)$ exponent conditioning

This subsection supplies external evidence using the CMU Pronouncing Dictionary (CMUdict) as an external pronunciation lexicon. The analysis isolates monosyllabic word types whose nucleus is AY (vowel-count = 1), treating AY as $\tau(A \rightarrow I)$. Orthographic exponent classes are defined by simple surface diagnostics (-y, i_e, igh, ie, other). The key test is environmental conditioning: if exponent choice is many-to-one, then exponent classes should covary with phonological environment (e.g., open vs closed codas) while preserving the same nucleus type $\tau(A \rightarrow I)$.

Table 24 — Exponent class by coda environment for monosyllabic $\tau(A \rightarrow I)$ types in CMUdict (n=949).

Exponent class	Closed (coda present)	Open (coda empty)	Total
i_e	237	0	237
ie-final	0	9	9
igh	40	6	46
other	565	60	625
y-final	0	32	32
Total	842	107	949

Figure 17 — Orthographic exponent distribution for monosyllabic $\tau(A \rightarrow I)$ (AY) types in CMUdict.



A small logistic model was fitted on the open-syllable subset (coda empty) to predict whether an AY type takes -y as its exponent as a function of onset complexity and surface word length. In this subset (n=107), -y exponents are concentrated in short forms with complex onsets, consistent with an exponent system rather than a change in nucleus type.

Table 25 — Logistic coefficients predicting -y exponent among open-syllable $\tau(A \rightarrow I)$ types (CMUdict).

Term	Coefficient (log-odds)	Interpretation (within open $\tau(A \rightarrow I)$ set)
Intercept	2.453	baseline log-odds of -y exponent
onset_size	1.895	more onset consonants $\rightarrow$ higher -y odds
word_len	-1.908	longer forms $\rightarrow$ lower -y odds
onset_first=DH	0.673	onset category effect

onset_first=SH	0.619	onset category effect
onset_first=S	0.501	onset category effect
onset_first=N	-0.332	onset category effect
onset_first=K	-0.335	onset category effect

These corpus patterns support the orthography hypothesis: exponent choice varies with environment while the nucleus type remains stable. On this view, -y is an exponent favored in short open-syllable forms, whereas i_e and igh are favored in closed environments. The mapping is therefore many-to-one, and segmental interpretations of Y are not required for nucleus-internal representation.

10E. Related work: not equivalent to tiers, empty nuclei, or gestures

Autosegmental decompositions. Autosegmental phonology makes directionality available only by sequencing or by adding autosegmental association conventions. A trajectory nucleus differs in object type: direction is internal to τ and does not require a second timing unit or an additional tier to represent movement. Tier-based decompositions can simulate direction but typically do so indirectly and at the cost of increasing representational degrees of freedom. Government/Strict-CV empty nuclei. Empty nucleus approaches explain distributional patterns by licensing and government relations over empty positions. The present analysis posits that diphthongal nuclei are filled by τ , shifting the explanatory center from licensing of empties to admissibility and markedness of trajectory objects. Licensing may remain relevant for consonantal patterns, but it is not the mechanism that generates nucleus directionality.

Gestural approaches. Gestural theories locate primitives in coordinated motor actions. The present analysis maintains a mapping to phonetic movement but denies ontological identity: τ is an abstract predicated transition, and physical movement is its realization. The commutative interface makes this separation explicit and testable.

Framework	Core ontology	How direction arises	Nucleus count expectation	Not-equivalent claim for τ -theory
Autosegmental tiers (V+G, VV)	Segments/features on tiers; direction from	Indirect	Often two objects or branching	τ is a single nucleus object with internal

	ordering/association			direction; no glide tier required
Government/Strict-CV empty nucleus	Empty positions licensed by government relations	Orthogonal (needs added content)	Empty category can count as slot	τ fills the nucleus; direction encoded in τ ; licensing not central for diphthongs
Gestural primitives	Motor actions as phonological primitives	Direct but ontologically motoric	Single gesture may be unit	Trajectory object is abstract; mapping to gesture constrained by commutative interface

Table 15 — Positioning: τ -theory is not equivalent to tier, empty-nucleus, or gestural ontologies.

10E.1 Sharp non-equivalence contrasts

Autosegmental decompositions: tiers can encode multiple linked targets, but direction and composition are not native without adding ordered relational structure that is isomorphic to τ .

Government/Strict-CV: empty nuclei explain absence conditions; τ explains the presence of a filled nucleus whose internal change is a licensed object, eliminating licensing as the explanatory center in diphthong regimes.

Gestural approaches: gestures model physical paths; τ is abstract predication constrained by commutativity, and therefore does not identify morphodynamical objects with motor programs.

10F. Executable commutativity: equation and space specification

The commutative interface can be stated as an equation that constrains mapping rather than serving as a rhetorical diagram.

Let $\llbracket \cdot \rrbracket \sqsubseteq$ map vowel states $v \in V$ into a phonetic manifold P (e.g., an acoustic $F1 \times F2$ space or a perceptual manifold), and let Φ map trajectory nuclei $\tau \in T$ into realized physical paths in a realization space A (e.g., articulatory or acoustic trajectories). Let R be a physical realization operator that takes two interpreted endpoints in P and returns a path in A .

Commutativity requires that realizing the operator-generated trajectory equals realizing the path computed from interpreted endpoints.

$$\Phi(Y(v_1, v_2)) = R(\llbracket v_1 \rrbracket \sqsubseteq, \llbracket v_2 \rrbracket \sqsubseteq)$$

$$\text{Domain: } v_1, v_2 \in V; Y(v_1, v_2) \in T; \llbracket \cdot \rrbracket \sqsubseteq : V \rightarrow P; \Phi : T \rightarrow A; R : P \times P \rightarrow A$$

Figure 13 — Executable commutativity constraint: realization commutes with predication (equation form).

11. Consequences and predictions

The ghost connector framework yields consequences that separate it from vowel-state and glide-tier analyses.

Prediction 1 (unit nucleus). Diphthongal nuclei should pattern as single units in syllable-sensitive processes when $\tau \in T$ is licensed by well geometry and template constraints.

Prediction 2 (template restriction). CVC templates can ban specific trajectories directly via $*CVC/\tau$ constraints rather than banning glides or vowel sequences.

Prediction 3 (orthography–phonology clustering). Y spellings should cluster where trajectory nuclei are licensed or favored, since Y is an exponent of predication rather than a vowel-state letter.

Prediction 4 (no licensing-head). Analyses that rely on empty nuclei or government relations should not treat Y as a governing/licensing head for diphthongs in the nucleus regime, because the nucleus position is filled by τ .

Prediction 5 (asymmetry). Where the well geometry is directional, asymmetries between $Y(v_1, v_2)$ and $Y(v_2, v_1)$ are expected in the admissible trajectory inventory and in repair behavior.

Prediction 6 (glide invisibility in nucleus regime). Processes that target onset glides should not target diphthongal nuclei as if they contained a glide object when the nucleus is τ .

12. Conclusion

Y is not a vowel, not a glide, and not an energy conduit. Y is a ghost connector: a silent predicator and directional operator that assigns transitions between vowel states inside a universal vowel well. In CVC structures, Y does not fill the nucleus slot. The nucleus slot is filled by a state nucleus in monophthongs and by a trajectory nucleus in diphthongs. English diphthongs are therefore predicated transitions, not blends. The mapping $Y(v_1, v_2)$ is not reducible to a plain feature bundle and does not require a glide tier. The vowel well is formalizable as a weighted directed graph $G = (V, E, w)$ with a partial operator Y defined exactly on admissible edges. A commutative interface links abstract predication to physical realization while preserving separation of levels and provides an executable constraint on mapping.

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Primary basis: morphodynamics documents on vowel wells and the Y operator (PDF sources in the current workspace).

Appendix A. Reported measurement tables and plots

Appendix A compiles the reported CVC measurement tables and reproduces their plots.

Rank	Vowel	Share (%)
1	o	37.21
2	a	29.07
3	e	18.6
4	i	7.92
5	u	7.2

Table 16 — Vowel-share ranking in English CVC nouns (reported).

C ₋₁ (given nucleus o)	Share (%)
f	45.64
n	27.62
h	5.94
t	4.44
l	3.77

Table 17 — Top C₋₁ distribution conditioned on nucleus o (reported).

Rank	CVC token (nucleus o)	Share (%)
1	for	45.38
2	not	20.14
3	now	4.68
4	how	4.37
5	top	3.7
6	non	1.5

7	low	1.41
8	job	1.36
9	box	1.29
10	hot	1.22

Table 18 — Top CVC tokens with nucleus o (reported).

Appendix B. Document-level term frequency diagnostics

Appendix B reports key term counts extracted from the morphodynamics corpus text.

Term	Count in uploaded text
Y	4947
vowel	2069
CVC	1343
well	1003
operator	722
energy	613
trajectory	552
depth	550
nucleus	489
lexical	335
ghost	325
glide	281
state	280
connector	271
predicator	109
diphthong	90
markedness	87
phonetic	57
conduit	56
transition	38
faithfulness	23

gesture	14
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Table 19 — Term counts extracted from the uploaded morphodynamics corpus text.

Appendix C. Document-attested y-containing lexicon (top 200)

Appendix C lists the top 200 alphabetic tokens containing the letter y in the uploaded corpus text (excluding second-person pronouns).

Token (contains y)	Frequency in uploaded text
theory	622
energy	613
trajectory	552
ontology	544
morphodynamical	408
system	400
phonology	347
category	316
physical	282
they	231
only	220
stability	188
systems	178
symbolic	174
articulatory	163
dynamical	157
syllable	151
style	150
morphodynamics	139
geometry	131
syntactic	117
synchrony	107
exactly	105

dynamics	103
frequency	98
structurally	92
dynamic	90
glyph	89
underlying	86
symbol	81
analysis	78
lexically	76
orthography	72
occupy	71
ready	67
syntax	67
harmony	55
says	52
fully	49
naturally	48
identity	46
empty	45
cyst	44
lynch	44
syllables	43
absolutely	42
merely	42
analyses	41
always	40
seamlessly	40
synchronies	40
inventory	39
already	36

everything	36
purely	36
arbitrary	35
crucially	34
commutativity	33
beyond	32
extremely	32
layer	32
morphology	32
optionally	32
prosody	32
slightly	32
boundary	31
typology	31
heavy	30
reality	30
reanalysis	30
directionality	29
energetically	28
infinity	28
longevity	28
physically	28
traditionally	28
symbols	26
especially	25
mathematically	25
type	25
ability	24
cleanly	24
conveys	24

many	24
monosyllables	24
morphodynamically	24
myth	24
property	24
sonority	24
summary	24
typological	23
occupying	22
scholarly	22
actually	20
clarify	20
density	20
entirely	20
functionally	20
happy	20
hierarchy	20
polyvalent	20
possibility	20
simply	20
sync	20
explanatory	19
necessity	19
overlay	19
carry	18
differently	18
novelty	18
reanalyzed	18
yields	18
flexibility	17

quality	17
study	17
systematically	17
city	16
compatibility	16
explicitly	16
optimality	16
orthographically	16
precisely	16
contemporary	15
dependency	15
fundamentally	15
very	15
vocabulary	15
away	14
cycle	14
directly	14
every	14
genuinely	14
internally	14
pathways	14
learnability	13
metaphysics	13
phonologically	13
sharply	13
strongly	13
theoretically	13
asymmetry	12
effectively	12
layers	12

literacy	12
lynx	12
productivity	12
rarely	12
universality	12
adequacy	11
compositionality	11
diachrony	11
entity	11
physics	11
clarity	10
continuity	10
literally	10
machinery	10
typing	10
applying	9
highly	9
increasingly	9
substantially	9
binary	8
categorically	8
chomsky	8
clearly	8
completely	8
consistently	8
dictionary	8
funny	8
heavily	8
mainly	8
maximally	8

monosyllabic	8
normally	8
overwhelmingly	8
oxygen	8
philosophy	8
roughly	8
simultaneously	8
story	8
stylistic	8
synchronic	8
terminology	8
typically	8
formally	7
hybrid	7
impossibility	7
likely	7
messy	7
opacity	7
rationality	7
symmetric	7
types	7
universally	7
activity	6
anomaly	6
apply	6
auditory	6
carrying	6

Table 20 — Top 200 y-containing tokens in the uploaded corpus text (frequency-ranked).